

CPTG Dynamics of ZF-COS-20115

Resolved Nonaxisymmetric Field and Orbit Closure

Carter L Glass Jr

Abstract

ZF-COS-20115 (Jekyll) is a compact massive quiescent galaxy observed at $z = 3.7135 \pm 0.0002$. Curvature Polarization Transport Gravity (CPTG) treats the observed baryonic distribution as the material source and describes the additional gravitational response through curvature polarization and finite-scale transport. Earlier CPTG work on this object was limited to an integrated effective-radius measurement; the resolved JWST stellar kinematics now permit a two-dimensional dynamical closure test. In the resolved reduction used here, each of 26 stellar-kinematic bins supplies its own structure-dependent acceleration $a_{\star,i}$ from its measured ordered and random support; $a_{\star}$ is not a universal acceleration constant. The same CPTG galaxy response law is then evaluated independently at each bin on a nonaxisymmetric stellar source, producing 26 local field vectors that are assembled into one conservative three-dimensional potential. The constructed conservative field realizes those 26 target vectors with maximum fractional numerical mismatch 3.66×10^{-12} . A 4096-trajectory orbit library is fully energy qualified and admits a nonnegative stationary superposition for which all 26 measured line-of-sight velocities and second moments lie within the specified marginal per-bin 3σ closure criterion at the strictest tabulated 1 percent tracer tolerance; the largest pulls are 2.431σ and 2.879σ . A matched 4096-trajectory baryonic-only control, using the same source geometry, orbit initial conditions, integration schedule, spatial convolution, spaxel operator, and closure solver, does not achieve all-bin 3σ second-moment closure: at 1 percent its maximum V_{rms} pull is 3.647σ , and it remains 3.072σ even at the loosest tabulated 10 percent tracer tolerance. Because the resolved kinematics enter both the local structural measurements and the final orbit weights, the CPTG result is a conditional dynamical compatibility and stationary-existence test rather than a blind prediction of the same data. Structural Mode is evaluated only after the field solution. The adopted sampled estimator gives $R_{95} = 8.19113 \text{ kpc}$, $\lambda = 2.38569 \text{ kpc}$, and $N = 3.43345$, remaining within 0.50% of the earlier oblate value $N = 3.45081$. Redshift fixes the observed CPTG phase, while the registered imaging and kinematic products supply the physical source coordinates; neither operation rescales the galaxy's structural acceleration.

1 Introduction

ZF-COS-20115, also known as ZF-COSMOS-20115 and Jekyll, was established as a compact massive quiescent system at $z \simeq 3.7$ [3, 4, 2]. The resolved JWST analysis gives the systemic redshift $z = 3.7135 \pm 0.0002$ [1]. Its large stellar mass, compact structure, early formation history, and sustained quiescence make it a useful test of any gravity framework that claims to connect baryonic structure to the field that supports a galaxy. The JWST/NIRSpec integral-field observations are especially valuable because they resolve the stellar kinematics across the galaxy rather than reducing the object to one integrated velocity dispersion [1].

Curvature Polarization Transport Gravity (CPTG) is formulated as an action-based nonlinear geometric theory in which ordinary baryonic stress-energy remains the material source while the gravitational response contains curvature-polarization and curvature-transport sectors [7]. In the quasi-static galaxy reduction used here, the measured baryonic field is mapped to the total gravitational response through an object-dependent structural acceleration, a leading polarization term, a finite transport scale, and nonlinear high-acceleration suppression. No additional collisionless matter source is introduced in this bound-system calculation. Structural Mode N is not a field variable or a fit parameter; it is evaluated only after the gravitational field has been solved and measures the radial organization of that solved field [10].

The earlier CPTG study of ZF-COS-20115 deliberately stopped at an integrated effective-radius reduction [6]. It used the available one-radius support measurement, obtained a high Structural Mode, $N_{\text{obl}} = 3.45081$, and identified resolved stellar kinematics as the natural next test. The present paper performs that test. The important advance is not the introduction of another fitted parameter. It is the replacement of one integrated structural measurement by 26 resolved structural measurements, followed by a single nonaxisymmetric field and orbit calculation.

The calculation proceeds in the same physical order as the theory. Redshift first places the observed galaxy on the CPTG native phase, while the registered imaging and kinematic products supply the angular-to-physical coordinate scale used by the numerical reconstruction. The resolved stellar kinematics then determine the local structural acceleration in each bin. The stellar light distribution supplies the nonaxisymmetric baryonic source. Those two ingredients are passed through the CPTG galaxy-scale response to obtain the resolved gravitational field. Because stellar orbits require a continuous conservative force rather than 26 isolated vectors, the field measurements are then embedded in one scalar potential. Only after that potential is fixed are orbit weights solved. Finally, Structural Mode is calculated from the full radial field as a downstream structural readout.

This ordering may be summarized as

$$\begin{aligned}
z &\longrightarrow \text{observed phase,} \\
\text{registered imaging/kinematic geometry} &\longrightarrow \text{physical coordinate scale,} \\
(V_i, \sigma_i, R_i) &\longrightarrow a_{\star,i}, \\
\{a_{\star,i}, \mathbf{g}_{\text{bar},i}\} &\longrightarrow \mathbf{g}_{\text{CPTG},i}, \\
\{\mathbf{g}_{\text{CPTG},i}\}_{i=1}^{26} &\longrightarrow \Phi_{\text{CPTG}}(\mathbf{x}) \longrightarrow \text{stationary orbit closure,} \\
g_{\text{CPTG}}(r) &\longrightarrow \lambda \longrightarrow N.
\end{aligned} \tag{1}$$

The sections below follow this same sequence so that the observational inputs, CPTG laws, numerical construction, and downstream interpretation remain visibly separated.

2 CPTG Galaxy-Scale Reduction

CPTG is defined from a variational extension of the Einstein–Hilbert action. In the convention of the parent theory, the master action is [7]

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + \mathcal{L}_{\text{m}} + \beta_A a_{\star}^{2/3} C^{1/3} + \mathcal{L}_{\text{trans}} \right], \tag{2}$$

The action is reproduced in the unit convention of the parent theory; dimensional galaxy-scale expressions below write factors of c explicitly where they enter the registered reduction. Here $\mathcal{L}_{\text{m}}$ is the ordinary matter Lagrangian, β_A is the action-level polarization coupling, $a_{\star}$ is the structure-dependent acceleration scale, C is the curvature-polarization invariant, and $\mathcal{L}_{\text{trans}}$ denotes the transport sector. In the static weak-field reduction relevant to galaxy dynamics, the corresponding representatives are [7]

$$\mathcal{L}_{\text{N}} = -\frac{|\nabla\Phi|^2}{8\pi G} - \rho\Phi, \quad \mathbf{g} = -\nabla\Phi, \tag{3}$$

and

$$C = g_j g_j + \kappa^2 (\partial_i g_j)(\partial_i g_j), \tag{4}$$

with polarization density

$$\mathcal{L}_{\text{pol}} = \beta_A a_{\star}^{2/3} C^{1/3}. \tag{5}$$

The Einstein–Hilbert term and ordinary matter source are therefore retained; the additional response is assigned to the geometric polarization and transport sectors rather than to an added collisionless matter component. The full covariant variation is not rederived in this application. The calculation below uses the established quasi-static galaxy reduction of the parent action and tests that reduction against the resolved stellar dynamics of ZF-COS-20115.

2.1 Structural acceleration $a_\star$

The role of $a_\star$ is central to the galaxy reduction and is distinct from the standard MOND acceleration scale a_0 [5]. In MOND, a_0 is taken to be universal and sets the acceleration scale of the modified response. CPTG does not assign that role to $a_\star$. The quantity $a_\star$ belongs to the structural state of the gravitating system or measurement element being reduced. It is therefore not universal, it is not fixed a priori across galaxies, and it is not identified with the solved local gravitational acceleration g .

Within the CPTG galaxy reduction, $a_\star$ sets the polarization amplitude and determines the derived finite transport radius $R_c = 48\pi c^2/a_\star$. Its numerical determination is reduction-dependent. In the resolved mixed-support construction used here, the measurement object is an individual stellar-kinematic bin, so each bin supplies

$$a_{\star,i} = \frac{V_i^2 + \sigma_i^2}{R_i} = \frac{V_{\text{rms},i}^2}{R_i}. \quad (6)$$

The 26 values of $a_{\star,i}$ are therefore structural measurements obtained from the resolved kinematic support before the CPTG field and orbit-weight solves; they are not 26 independently adjusted gravity parameters. This distinction is essential to the interpretation of the calculation.

2.2 Reduced galaxy response

In the quasi-static galaxy reduction, the observed baryonic matter supplies the material source and g_{bar} denotes its gravitational field. Once g_{bar} and the structural state appropriate to the measurement are specified, the reduced CPTG law determines the local gravitational response [7, 9].

The first additional term is the curvature-polarization response,

$$g_{\text{pol}} = \sqrt{a_\star g_{\text{bar}}}. \quad (7)$$

This term couples the baryonic source strength to the structural acceleration of the measured system. The same structural scale defines the finite curvature-transport radius

$$R_c = \frac{48\pi c^2}{a_\star}. \quad (8)$$

At finite radius r , the corresponding geometric transport contribution is

$$g_{\text{geom}}(r) = \frac{\frac{1}{2}a_\star(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \quad (9)$$

The local response is nonlinear. CPTG regulates the polarization-plus-transport contribution through

$$\Sigma_g = \left[1 + \left(\frac{g}{a_\star} \right)^{123/50} \right]^{-11/50}, \quad (10)$$

so that the solved field satisfies

$$g = g_{\text{bar}} + (g_{\text{pol}} + g_{\text{geom}})\Sigma_g. \quad (11)$$

Equation (11) is implicit because the suppression factor depends on the solved field itself. For $g_{\text{bar}} \geq 0$ and nonnegative polarization-plus-transport amplitude, the residual function is strictly increasing in g ; the admissible local solution is therefore unique and nonnegative [7]. The baryonic source, structural acceleration, polarization response, finite transport contribution, and nonlinear suppression are thus distinct terms within one solved galaxy reduction rather than independently fitted components.

For a system supported by both ordered and random stellar motion, the observed second moment is

$$V_{\text{rms}}^2 = V^2 + \sigma^2. \quad (12)$$

The parent CPTG framework distinguishes such an observed support coordinate from a universal definition of a_* ; the identification of support with structural acceleration belongs to the resolved reduction adopted here [7]. In this reduction each spatial bin is treated as an independent measurement object for the galaxy-scale law. Its measured V , σ , and structural radius determine its own $a_{*,i}$, and that structural state is then passed through Eqs. (7)–(11). The analogy to a SPARC application is therefore at the level of an independently evaluated measurement object, not at the level of imposing one common structural acceleration across the 26 bins of this two-dimensional stellar field.

With the galaxy law defined, the next step is to place the observed JWST measurements on the physical coordinate system used by the bound-system calculation.

3 Observed Epoch, Resolved Data, and Physical Coordinates

3.1 Redshift as coordinate placement

The resolved JWST analysis gives the systemic redshift $z = 3.7135 \pm 0.0002$ [1]. At the central value,

$$a = \frac{1}{1+z} = 0.2121565715, \quad N_\pi = \ln a = -1.550430732. \quad (13)$$

As in the preceding ZF-COS-20115 analysis, the acoustic and luminosity CPTG distance coordinates are used only to express the observed source in physical units; they are projections of the same native phase rather than separate galaxy histories [8, 6]. The processed source and observation products used by the field-to-orbit calculation carry the fixed physical registration

$$1 \text{ arcsec} = 7.09146261 \text{ kpc}, \quad 0.03 \text{ arcsec} = 0.21274388 \text{ kpc}. \quad (14)$$

This source-coordinate registration is used consistently for the resolved bin positions, spaxel support, source reconstruction, and orbit observation operator. The orbit calculation is therefore performed entirely in one fixed physical coordinate system.

The systemic redshift fixes the observed CPTG phase, while the processed source products carry the angular-to-physical registration used by the bound-system calculation. No factor of $H(z)$ or $E(z)$ multiplies or divides the measured structural acceleration. In particular,

$$a_{\star,i} \not\propto E(z)a_{\star,i}, \quad a_{\star,i} \not\propto \frac{a_{\star,i}}{E(z)}. \quad (15)$$

The epoch and physical source registration therefore enter as coordinate information; the galaxy’s measured stellar support supplies the structural state.

3.2 Resolved stellar kinematics

The resolved analysis uses 26 stellar-kinematic bins supported by 141 exact spaxels. The public JWST/NIRSpec observations of Jekyll were obtained in programme 1217; the spatially resolved spectroscopy and reduction context are described by Pérez-González et al. [1]. The numerical evidence used here begins from the processed bin velocities, dispersions, uncertainties, and exact spaxel assignments rather than from raw detector exposures.

For every bin,

$$V_{\text{rms},i} = \sqrt{V_i^2 + \sigma_i^2}. \quad (16)$$

The adopted resolved structural radius is

$$R_i = \sqrt{x_i^2 + \left(\frac{z_i}{0.70}\right)^2}. \quad (17)$$

The choice $q_{\text{str}} = 0.70$ follows the $\epsilon = 0.30$ elliptical geometry used for the resolved stellar annuli in the JWST analysis [1]. It is distinct from the projected axis ratio of the reconstructed stellar source introduced below; the two quantities have different roles and are not interchanged. The sensitivity of the inferred structural acceleration to this radius convention is retained explicitly. Replacing $q_{\text{str}} = 0.70$ by the source projected axis ratio 0.824798 changes $a_{\star,i}$ by a median absolute fraction 0.0954 and a maximum 0.1767; using a circular radius changes it by a median absolute fraction 0.1987 and a maximum 0.4234. The $q_{\text{str}} = 0.70$ geometry is therefore a substantive part of the resolved measurement interface rather than an innocuous plotting convention.

The fixed physical-coordinate registration and the adopted support geometry supply the structural radius of every bin, while the spectroscopy supplies its ordered and random stellar support. These are the quantities needed to make the resolved structural measurement.

4 Local Structural Acceleration

For a resolved mixed-support measurement, the local stellar second moment quantifies the combined ordered and random support at that location. Following the measurement interface defined in Sec. 2, the structural acceleration is

$$a_{\star,i} = \frac{V_i^2 + \sigma_i^2}{R_i} = \frac{V_{\text{rms},i}^2}{R_i}. \quad (18)$$

Equation (18) is the resolved realization of the structure-dependent CPTG quantity introduced above. It does not make $a_{\star}$ a universal constant, a MOND-like transition acceleration, or the pointwise gravitational field. The observational unit is the resolved bin, so each bin is treated as the measurement object for the local reduction and evaluated independently under the same CPTG galaxy law. The 26 values are not collapsed to a galaxy-wide $a_{\star}$, and no later structural diagnostic is allowed to select or modify any $a_{\star,i}$.

The measured values span

$$5.42746 \times 10^{-10} \leq a_{\star,i} \leq 2.25480 \times 10^{-8} \text{ m s}^{-2}, \quad (19)$$

with median

$$\tilde{a}_{\star} = 4.53765 \times 10^{-9} \text{ m s}^{-2}. \quad (20)$$

The quoted V_{rms} uncertainties imply substantial uncertainty in the structural inputs themselves. Linear propagation through $a_{\star,i} = V_{\text{rms},i}^2/R_i$, holding the adopted structural radii fixed, gives fractional 1σ uncertainties from 0.207 to 0.686, with median 0.314. A 200,000-draw per-bin Monte Carlo propagation of the measured V_{rms} uncertainties through the local CPTG response, with R_i and $g_{\text{bar},i}$ fixed, gives central 68-percent fractional half-widths in $g_{\text{CPTG},i}$ from 0.0235 to 0.194, with median 0.0788. These numbers quantify the uncertainty of the local structural-input map only; they do not marginalize the continuous-field reconstruction or orbit-weight solution.

Bin 21, which has the minimum local $a_{\star}$, is retained without clipping, down-weighting, or special treatment.

Table 1: Resolved structural measurements. Accelerations in the final column are in units of 10^{-9} m s^{-2} .

Bin	R_i (kpc)	V_i (km s $^{-1}$)	σ_i (km s $^{-1}$)	$V_{\text{rms},i}$ (km s $^{-1}$)	$a_{\star,i}$
0	0.1873	-12.52	331.09	331.33	18.9937
1	0.0894	44.00	245.45	249.36	22.5480
2	0.3658	49.54	259.83	264.51	6.1992
3	0.3465	21.31	271.99	272.82	6.9611
4	0.6522	28.07	220.66	222.44	2.4586
5	0.4257	73.89	217.53	229.74	4.0179
6	1.3459	-101.72	239.35	260.07	1.6287
7	0.9134	-72.25	238.19	248.90	2.1980
8	0.7086	-3.49	315.05	315.07	4.5400

Continued on next page

Table 1 continued from previous page

Bin	R_i (kpc)	V_i (km s $^{-1}$)	σ_i (km s $^{-1}$)	$V_{\text{rms},i}$ (km s $^{-1}$)	$a_{\star,i}$
9	1.4411	91.29	252.43	268.43	1.6203
10	0.6263	-8.14	347.25	347.34	6.2430
11	1.1776	6.12	454.08	454.12	5.6753
12	0.5382	-24.95	412.56	413.32	10.2873
13	1.1995	-42.75	407.47	409.71	4.5353
14	0.6535	-53.11	396.72	400.26	7.9456
15	0.3336	4.06	350.30	350.32	11.9232
16	0.2083	50.24	282.89	287.32	12.8453
17	0.4568	42.32	234.32	238.11	4.0224
18	0.6677	87.18	424.08	432.94	9.0983
19	1.3207	48.95	388.34	391.41	3.7594
20	0.5685	39.16	303.44	305.96	5.3368
21	0.8152	90.38	74.05	116.84	0.5427
22	0.9213	9.31	280.58	280.73	2.7722
23	1.3409	79.47	253.31	265.48	1.7034
24	1.2959	14.03	160.87	161.48	0.6521
25	1.3639	144.66	201.92	248.39	1.4660

The resolved structural measurements are now fixed. The next ingredient is independent of those kinematic values: the baryonic source field reconstructed from the stellar light.

5 Nonaxisymmetric Stellar Source

CPTG does not infer the material source from the kinematic residuals. The baryonic field is constructed separately from the observed stellar distribution, and only afterward is the geometric response evaluated. For ZF-COS-20115 the rest-frame optical F444W light is used to reconstruct a nonaxisymmetric stellar source rather than forcing the galaxy into a spherical or purely axisymmetric approximation.

The stellar source is represented by 30 nested oblate components extending to 15.574 kpc. Their total modeled mass is

$$M_{\star,\text{model}} = 1.57036301578 \times 10^{11} M_{\odot}. \quad (21)$$

This extended model total should not be compared directly with a finite-aperture stellar mass. Evaluating the same nested source inside a 3 kpc semimajor-axis ellipse using the source model's own projected axis ratio gives

$$M_{\star,\text{model}}(< 3 \text{ kpc}) = 1.15496 \times 10^{11} M_{\odot}. \quad (22)$$

This is compared with the JWST stellar-mass-map measurement $M_{\star}(< 3 \text{ kpc}) = (1.16 \pm 0.15) \times 10^{11} M_{\odot}$ reported for the observational 3 kpc semimajor-axis aperture [1]; the two aperture geometries are close but are not assumed to be identical. The projected source axis ratio of the adopted component reconstruction is $q_{\text{src,proj}} = 0.824797606$. For the

three-dimensional lifting used in the field calculation, the intrinsic axis ratio is set equal to the projected value,

$$q_{\text{src,int}} = q_{\text{src,proj}} = 0.824797606. \quad (23)$$

This equality is an adopted deprojection/lifting assumption of the source model, not an independent measurement of the intrinsic thickness. The nested components describe the radial mass profile, while the reconstructed departure from perfect axisymmetry is represented by a centroid-conserving first azimuthal harmonic. In elliptical radius s and azimuth θ , the surface-density modulation is

$$\Sigma(s, \theta) = \Sigma_0(s) \left[1 + A_1 \left(1 - \frac{s}{R_{\text{node}}} \right) \cos(\theta - \phi_1) \right], \quad (24)$$

where A_1 controls the strength of the asymmetry, R_{node} is the radius at which the harmonic changes sign, and ϕ_1 sets its orientation. The source-reconstruction parameters used by the field calculation are

$$A_1 = 0.302170605, \quad (25)$$

$$R_{\text{node}} = 5.59359355 \text{ kpc}, \quad (26)$$

$$\phi_1 = -47.02235^\circ \quad (27)$$

in elliptical coordinates and sky position angle 158.42789° . The projected harmonic factor is held constant along the line of sight in the three-dimensional source perturbation. These parameters belong to the independently constructed source model and are not adjusted in the orbit-weight closure. The adopted source realization remains positive throughout the modeled spatial support.

The nonaxisymmetric perturbation is evaluated with deterministic $38 \times 48 \times 128$ ellipsoidal quadrature. The axisymmetric source field and scalar potential are evaluated directly with fixed 96-node Gauss quadrature. A 0.01 kpc field regularization is retained in the source evaluation. This construction supplies the baryonic acceleration vector at every resolved bin without using the stellar velocities to change the source. We can now combine the independently measured structural state with the independently reconstructed baryonic field.

6 Resolved CPTG Field

At this stage the two required inputs are available independently: each bin has a measured structural acceleration $a_{\star,i}$, and the stellar reconstruction supplies a nonaxisymmetric baryonic vector $\mathbf{g}_{\text{bar},i}$. The CPTG galaxy law can therefore be evaluated without changing either input.

Let

$$g_{\text{bar},i} = |\mathbf{g}_{\text{bar},i}|. \quad (28)$$

For each bin, the polarization amplitude is

$$g_{\text{pol},i} = \sqrt{a_{\star,i} g_{\text{bar},i}}, \quad (29)$$

the structural acceleration fixes its finite transport radius,

$$R_{c,i} = \frac{48\pi c^2}{a_{\star,i}}, \quad (30)$$

and the finite-radius geometric contribution is

$$g_{\text{geom},i} = \frac{\frac{1}{2}a_{\star,i}(R_i/R_{c,i})^{7/5}}{1 + (R_i/R_{c,i})^{12/5}}. \quad (31)$$

The nonlinear suppression factor is

$$\Sigma_{g,i} = \left[1 + \left(\frac{g_i}{a_{\star,i}} \right)^{123/50} \right]^{-11/50}, \quad (32)$$

and the positive local solution satisfies

$$g_i = g_{\text{bar},i} + (g_{\text{pol},i} + g_{\text{geom},i})\Sigma_{g,i}. \quad (33)$$

Thus the same CPTG response law is solved 26 times, once for each measured structural state. No cross-bin fit is used to choose the individual $a_{\star,i}$ values.

The scalar solution gives the local response magnitude. The nonaxisymmetric source calculation already supplies the local direction, so the resolved CPTG target vector is

$$\mathbf{g}_{\text{CPTG},i} = \frac{g_i}{g_{\text{bar},i}} \mathbf{g}_{\text{bar},i}. \quad (34)$$

This preserves the locally computed baryonic-field direction while changing its magnitude according to the CPTG response. The result is a set of 26 resolved gravitational vectors. They are the field measurements to be realized dynamically; Structural Mode has not entered the construction.

A set of isolated vectors is sufficient to state the resolved field at the measured locations, but stellar trajectories require a continuous conservative force everywhere between those locations. The next step is therefore a constrained conservative reconstruction: construct one scalar potential whose gradient realizes the resolved CPTG field at all 26 anchors.

7 Conservative Three-Dimensional Potential

The resolved CPTG vectors define the gravitational field where the stellar kinematics are measured. To test whether those local measurements can belong to one stationary galaxy, they must be connected by a single conservative three-dimensional field. Orbit integration therefore requires one scalar potential whose gradient realizes the resolved vector field. The base potential is the direct nested-oblate potential of the axisymmetric component of the stellar source. A curl-free Hermite Gaussian correction is then added to realize, at the 26 anchors, the difference between that base field and the full nonaxisymmetric CPTG target. The correction therefore carries both the nonaxisymmetric source contribution not represented by the base component and the CPTG response required by Eq. (34). If the i th correction center is at $(x_i, 0, z_i)$, with width h_i , coefficient vector $(c_{x,i}, c_{z,i})$, and

$$d_i^2 = (x - x_i)^2 + y^2 + (z - z_i)^2, \quad (35)$$

then the correction potential is

$$\Phi_H(\mathbf{x}) = - \sum_i \frac{(x - x_i)c_{x,i} + (z - z_i)c_{z,i}}{h_i^2} \exp\left(-\frac{d_i^2}{2h_i^2}\right). \quad (36)$$

The complete orbit potential is

$$\Phi_{\text{CPTG}} = \Phi_{\text{source}} + \Phi_H, \quad \mathbf{g}_{\text{orbit}} = -\nabla\Phi_{\text{CPTG}}. \quad (37)$$

At all 26 structural centers, the actual orbit field is compared directly with Eq. (34). The median fractional vector mismatch is

$$3.8137 \times 10^{-13}, \quad (38)$$

and the maximum is

$$3.6611 \times 10^{-12}, \quad (39)$$

well below the 10^{-3} field-realization criterion. Separate three-dimensional source-quadrature spot checks give maximum force and potential mismatches 5.73×10^{-12} and 3.61×10^{-15} , respectively.

Exact agreement at the 26 centers is an interpolation constraint rather than an independent prediction between centers. A leave-one-center-out diagnostic gives median held-out fractional mismatch 0.290 and maximum 1.052 at bin 24; the corresponding held-out direction mismatch at bin 24 is 165.1° . Bins 21 and 23 also show substantial held-out sensitivity, with fractional mismatches 0.779 and 0.769 and direction mismatches 40.9° and 38.8° , respectively. The 26 anchors therefore do not uniquely determine the continuous field between them. The Hermite construction used below is one conservative realization satisfying all supplied anchors, and the dynamical conclusion is restricted to that realization; robustness across alternative interpolation rules is not claimed. With the conservative field fixed, the problem changes from field construction to stellar dynamics: can a stationary population of orbits in this potential reproduce the resolved observations?

8 Orbit Library and Observation Operator

The conservative potential is now held fixed. The orbit stage does not alter the CPTG field; it asks whether that field can support a nonnegative stationary stellar population with the observed projected kinematics. We use the standard orbit-superposition idea introduced by Schwarzschild: integrate many representative stellar trajectories in one fixed potential, project each trajectory into the same observables measured on the sky, and then ask whether nonnegative weights can combine those trajectories into the observed galaxy [11, 12]. The orbit library contains 4096 initial phase-space states. Each orbit is integrated in the same conservative potential for 12 local periods using DOP853 with relative tolerance 10^{-9} and absolute tolerance 10^{-11} . The qualification requirement is

$$\frac{\max_t |E(t) - E(0)|}{|E(0)|} \leq 5 \times 10^{-4}. \quad (40)$$

All 4096 trajectories satisfy the energy criterion. The largest fractional energy error is

$$4.59439 \times 10^{-5}, \quad (41)$$

with no integration retries. Uniform observation sampling uses 4096 times on $[0, T)$ for each trajectory.

The sky plane is (x, z) and the line of sight is the y axis. Let k index an orbit, n its uniform time sample, and j an observed spaxel with sky position (x_j, z_j) . Then $v_{\text{los}, kn} = v_{y, kn}$ and

$$d_{\text{sky}, knj}^2 = (x_{kn} - x_j)^2 + (z_{kn} - z_j)^2, \quad (42)$$

with unnormalized Gaussian observation weight

$$W_{knj} = \exp \left[-\frac{d_{\text{sky}, knj}^2}{2\sigma_{\text{PSF}}^2} \right]. \quad (43)$$

The adopted point-spread-function scale is

$$\sigma_{\text{PSF}} = 0.247200 \text{ kpc}, \quad \text{FWHM} = 0.582112 \text{ kpc} = 0.0820862 \text{ arcsec}. \quad (44)$$

Let $\mathcal{S}_i$ denote the exact set of spaxels assigned to resolved bin i , and let $\langle \cdot \rangle_n$ denote the uniform average over the 4096 time samples of orbit k . The orbit-to-bin moment matrices are then

$$L_{ki} = \left\langle \sum_{j \in \mathcal{S}_i} W_{knj} \right\rangle_n, \quad (45)$$

$$M_{1, ki} = \left\langle \sum_{j \in \mathcal{S}_i} W_{knj} v_{\text{los}, kn} \right\rangle_n, \quad (46)$$

$$M_{2, ki} = \left\langle \sum_{j \in \mathcal{S}_i} W_{knj} v_{\text{los}, kn}^2 \right\rangle_n. \quad (47)$$

A 2048-versus-4096 temporal-sampling comparison on 32 trajectories gives maximum changes below 0.58 km s^{-1} in V and 0.40 km s^{-1} in V_{rms} . These moment matrices form the bridge between the continuous orbit library and the 26 observed bins. The remaining task is to determine whether nonnegative orbit weights can satisfy the tracer distribution and the measured first and second moments simultaneously.

9 Nonnegative Stationary Orbit Closure

Each integrated orbit contributes a known amount of tracer light, mean line-of-sight motion, and squared line-of-sight motion to each observed bin. The only quantities solved at this stage are the nonnegative orbit weights. No CPTG field parameter is adjusted. Let $\omega_k \geq 0$ be the orbit weights. For a hard tracer tolerance τ , the projected tracer fractions are constrained by

$$(1 - \tau) \ell_i \leq \sum_k \omega_k L_{ki} \leq (1 + \tau) \ell_i, \quad (48)$$

where ℓ_i is the observed tracer fraction. The first and second raw velocity moments are included in one linear minimax problem after normalization by their observational scales. The reconstructed observables are

$$V_i^{\text{model}} = \frac{\sum_k \omega_k M_{1,ki}}{\sum_k \omega_k L_{ki}}, \quad (49)$$

$$V_{\text{rms},i}^{\text{model}} = \left(\frac{\sum_k \omega_k M_{2,ki}}{\sum_k \omega_k L_{ki}} \right)^{1/2}. \quad (50)$$

The tabulated tracer tolerances are 1, 1.5, 2, 2.5, 3, 5, 7.5, and 10 percent. The strictest tabulated value, $\tau = 1\%$, gives all 26 velocity and all 26 second-moment marginal pulls below 3σ . The tolerance is a numerical tracer-closure tolerance, not an observational uncertainty. A finer diagnostic below one percent locates the transition more precisely: the CPTG solution has maximum $|P_{V_{\text{rms}}}| = 3.025$ at $\tau = 0.25\%$ and 2.949 at $\tau = 0.50\%$, while exact tracer equality ($\tau = 0$) gives 3.105σ . The matched baryonic-only control remains above 3σ throughout the same 0–1% diagnostic interval. Thus the reported 1% row is a convenient strict tabulated operating point, not an exact-tracer closure claim. The optimization objective is the minimax residual of the normalized raw moments subject to the tracer bounds, whereas the displayed maximum V_{rms} pull is formed afterward from the nonlinear ratio in Eq. (50). Consequently, relaxing τ need not make the reported maximum V_{rms} pull monotonic; the rise from the 7.5 percent to the 10 percent row is therefore not, by itself, a solver inconsistency.

Three solver algorithms available through SciPy/HiGHS—HiGHS, dual simplex, and interior point—return the same optimum at $\tau = 1\%$, with maximum linear-constraint violation 1.39×10^{-17} .

Tracer tol. (%)	Max $ P_V $	Bin	Max $ P_{V_{\text{rms}}} $	Bin	Active orbits	$< 3\sigma$
1.0	2.431	9	2.879	11	23	Yes
1.5	2.390	9	2.829	12	19	Yes
2.0	2.365	9	2.807	12	21	Yes
2.5	2.374	3	2.813	11	19	Yes
3.0	2.321	18	2.804	11	18	Yes
5.0	2.277	18	2.853	11	17	Yes
7.5	2.223	18	2.992	11	15	Yes
10.0	2.184	18	3.118	11	16	No

Table 2: Tracer-tolerance sensitivity. “Active orbits” counts weights greater than 10^{-10} . The 1 percent row is the strictest tabulated tolerance.

At $\tau = 1\%$,

$$\max_i |P_{V,i}| = 2.43077 \quad (\text{bin } 9), \quad (51)$$

$$\max_i |P_{V_{\text{rms}},i}| = 2.87873 \quad (\text{bin } 11). \quad (52)$$

There are 23 materially nonzero orbit weights above 10^{-10} . Bin 21 remains fully included and gives

$$P_{V,21} = -2.39844, \quad P_{V_{\text{rms}},21} = +2.06354. \quad (53)$$

Table 3: Resolved observational constraints and the nonnegative orbit-superposition closure at 1 percent tracer tolerance. Observed quantities include their quoted 1σ uncertainties.

Bin	V_{obs}	V_{model}	P_V	$V_{\text{rms,obs}}$	$V_{\text{rms,model}}$	$P_{V_{\text{rms}}}$
0	-12.5 ± 28.6	-41.5	-1.01	331.3 ± 38.3	298.3	-0.86
1	44.0 ± 25.1	-0.5	-1.78	249.4 ± 31.5	288.6	+1.25
2	49.5 ± 27.9	-11.4	-2.18	264.5 ± 35.0	279.5	+0.43
3	21.3 ± 29.0	-46.9	-2.36	272.8 ± 36.1	280.7	+0.22
4	28.1 ± 29.0	17.2	-0.38	222.4 ± 36.7	255.6	+0.90
5	73.9 ± 31.1	42.5	-1.01	229.7 ± 33.4	245.0	+0.46
6	-101.7 ± 49.8	-36.1	+1.32	260.1 ± 75.5	250.5	-0.13
7	-72.3 ± 37.8	-21.7	+1.34	248.9 ± 50.3	253.1	+0.08
8	-3.5 ± 38.0	-47.1	-1.15	315.1 ± 44.9	241.7	-1.63
9	91.3 ± 55.1	-42.7	-2.43	268.4 ± 74.8	218.2	-0.67
10	-8.1 ± 38.1	-63.0	-1.44	347.3 ± 53.4	254.1	-1.75
11	6.1 ± 80.4	17.1	+0.14	454.1 ± 82.9	215.6	-2.88
12	-25.0 ± 36.8	-56.1	-0.85	413.3 ± 42.8	290.3	-2.87
13	-42.7 ± 119.1	139.1	+1.53	409.7 ± 140.6	330.9	-0.56
14	-53.1 ± 37.8	-24.6	+0.75	400.3 ± 42.8	306.3	-2.20
15	4.1 ± 27.4	-25.4	-1.08	350.3 ± 36.4	307.1	-1.19
16	50.2 ± 24.1	12.5	-1.57	287.3 ± 34.2	292.1	+0.14
17	42.3 ± 29.2	59.3	+0.58	238.1 ± 36.6	230.8	-0.20

Continued on next page

Table 3 continued from previous page

Bin	V_{obs}	V_{model}	P_V	$V_{\text{rms,obs}}$	$V_{\text{rms,model}}$	$P_{V_{\text{rms}}}$
18	87.2 ± 34.6	3.4	-2.42	432.9 ± 45.4	303.9	-2.84
19	48.9 ± 82.3	113.7	+0.79	391.4 ± 94.8	302.2	-0.94
20	39.2 ± 36.1	51.4	+0.34	306.0 ± 48.9	258.1	-0.98
21	90.4 ± 18.6	45.7	-2.40	116.8 ± 22.5	163.3	+2.06
22	9.3 ± 40.0	52.6	+1.08	280.7 ± 55.1	264.8	-0.29
23	79.5 ± 63.6	-57.6	-2.16	265.5 ± 73.8	184.4	-1.10
24	14.0 ± 42.1	36.6	+0.54	161.5 ± 48.4	187.7	+0.54
25	144.7 ± 59.0	90.6	-0.92	248.4 ± 75.0	219.1	-0.39

The table demonstrates compatibility of the measured first and second moments with a stationary nonnegative distribution in the constructed field. It must not be read as 52 independent blind predictions: the same resolved kinematics determine the local $a_{\star,i}$ measurements and participate in the final orbit-weight optimization.

9.1 Matched baryonic-only dynamical control

A separate control calculation tests whether the orbit-superposition freedom by itself is sufficient to obtain the same resolved closure. The CPTG augmentation is removed while the measured nonaxisymmetric stellar source is retained. A conservative baryonic-only potential is constructed to match the 26 nonaxisymmetric baryonic field vectors, and the full dynamical calculation is then repeated with the same 4096 phase-space initial conditions, 12 local integration periods, 4096 uniform time samples, DOP853 tolerances, energy gate, PSF, exact-spaxel observation operator, nonnegative orbit weights, and tracer-tolerance ladder used for the CPTG calculation.

The baryonic control is numerically qualified independently. Its conservative field reproduces the 26 baryonic target vectors with maximum fractional mismatch 5.25×10^{-16} . All 4096 trajectories pass the energy gate, the maximum fractional energy error is 3.41×10^{-5} , and no integration retry is required.

At the 1 percent tracer tolerance, the comparison is

Quantity	Baryonic only	CPTG
Maximum $ P_V $	2.534	2.431
Maximum $ P_{V_{\text{rms}}} $	3.647	2.879
V_{rms} RMSE (km s^{-1})	105.35	75.30
V_{rms} MAE (km s^{-1})	76.55	54.96
Bins with $ P_{V_{\text{rms}}} < 3$	24/26	26/26

Table 4: Matched baryonic-only control and CPTG closure at the same 1 percent numerical tracer tolerance. The two calculations use the same resolved observations and the same downstream orbit and observation machinery; the gravitational field is the changed element.

The baryonic-only result does not satisfy the all-bin marginal 3σ second-moment closure criterion anywhere on the tested 1–10 percent tracer-tolerance ladder. Its smallest listed

maximum $|P_{V_{\text{rms}}}|$ is 3.072 at 10 percent. Within the common sampled phase-space basis, the control therefore shows that the CPTG closure is not reproduced merely by allowing the same nonnegative orbit library to adjust to the resolved kinematic constraints. This comparison remains a matched conditional-compatibility test rather than an out-of-sample prediction, because the orbit weights in both potentials are optimized against the measured first and second moments.

9.2 Radialized resolved-support diagnostic

A one-dimensional diagnostic is useful for visualizing the otherwise two-dimensional resolved solution. A direct sort of the 26 bins by radius would mix different azimuths and create an artificial sequence. The radial profile is therefore constructed with the same no-extrapolation geometry used for the two-dimensional structural reduction. The transformed coordinates are

$$u = x_{\text{major}}, \quad v = \frac{x_{\text{minor}}}{0.70}, \quad (54)$$

and the 26 bin centroids are triangulated in the (u, v) plane. At each fixed elliptical radius, the piecewise-linear field is averaged around the complete centered ellipse. The radial display is terminated at

$$R_{\text{complete}} = 1.081633314 \text{ kpc}, \quad (55)$$

the largest centered transformed circle lying wholly inside the observed convex hull. No angular or radial extrapolation enters this display.

The observed, baryonic-only, and CPTG second-moment fields are all passed through this identical linear radial operator. The black points in Figure 1 sample the radialized observed V_{rms} field with formal propagated uncertainties. The blue curve comes from the independently integrated baryonic-only orbit control described above, and the orange curve comes from the CPTG orbit-closure solution. Over the complete-coverage radial interval, the radialized V_{rms} RMSE is 50.38 km s^{-1} for the baryonic-only control and 37.96 km s^{-1} for CPTG. These values are descriptive summaries of the radialized display, not replacements for the underlying 26-bin closure statistics.

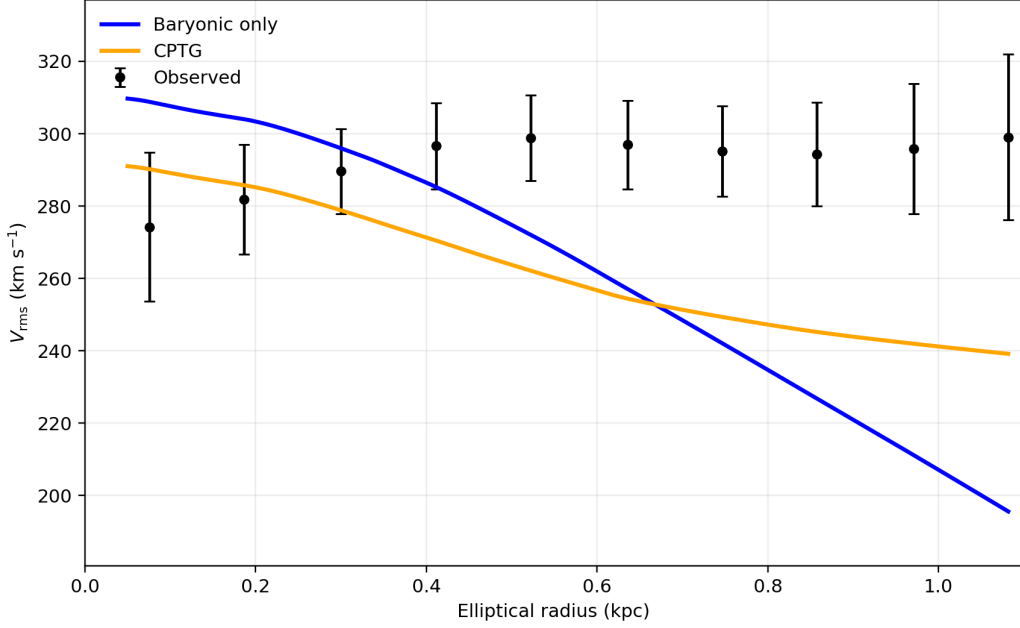


Figure 1: Resolved radial second-moment comparison for ZF-COS-20115. Black points with formal propagated uncertainties sample the observed two-dimensional V_{rms} field after complete-ellipse radialization. The blue curve is the identically radialized matched baryonic-only orbit solution, and the orange curve is the identically radialized CPTG orbit-closure solution. All three use the same two-dimensional-to-radial operator and terminate at $R_{\text{complete}} = 1.081633314$ kpc. The panel is a pressure-supported analogue of a rotation-curve comparison, not a literal circular-speed curve.

10 Full-Galaxy Structural Mode

Structural Mode N is a downstream structural coordinate, not a force-law term or orbit-fit parameter. It quantifies where the radial variation of the solved CPTG field is concentrated over the adopted outer radial interval [10]. A larger N corresponds to structural variation weighted farther inward; a smaller N places more of that variation toward the outer part of the interval. In this paper Structural Mode is evaluated only after the resolved field and dynamical closure have been established. It does not generate any $a_{\star,i}$, potential coefficient, orbit, or orbit weight.

The full radial continuation preserves all 26 measured inner structural accelerations. The resolved data end well inside the radius used by the Structural Mode measurement, so the field must be continued outward without replacing the measured inner solution. Outside the resolved footprint, the local structural acceleration is continued through the nonlinear spherical Jeans relation

$$\frac{dy}{dr} = -g_{\text{CPTG}}\left(r; a_{\star} = \frac{y}{r}\right) - y \frac{d \ln \nu}{dr} - \frac{2\beta_J y}{r}, \quad y = \sigma_r^2, \quad (56)$$

with fiducial $\beta_J = 0$. The supplied full- R_{95} continuation uses a finite-volume source interface and a shape-preserving radial interpolation. The reported Structural Mode is evaluated from that supplied continuation; the present archive independently verifies the resulting profile, sampled estimator, continuum cross-check, and radial-sampling convergence.

The adopted outer radius is

$$R_{95} = 8.191131986 \text{ kpc.} \quad (57)$$

Structural Mode begins from a scalar that records both field strength and radial structure,

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc.} \quad (58)$$

The first term carries the local field amplitude and the second carries the radial gradient on the adopted one-kiloparsec structural scale. The structural weight

$$w(r) = \left| \frac{dC_g}{dr} \right| \quad (59)$$

therefore marks the radii at which the solved field changes most strongly. Its weighted radial centroid is

$$\lambda = \frac{\int_{R_{95}/5}^{R_{95}} r w(r) dr}{\int_{R_{95}/5}^{R_{95}} w(r) dr}, \quad (60)$$

and the dimensionless Structural Mode is

$$N = \frac{R_{95}}{\lambda}. \quad (61)$$

Thus N does not measure the amount of gravitating mass or the quality of the orbit fit. It measures where the radial variation of the already solved field is concentrated relative to the adopted outer radius.

For direct comparability with the SPARC implementation, the reported value uses the same sampled estimator: numerical radial derivatives of g and C_g , the same five-sample smoothing of $|dC_g/dr|$, and the same $R_{95}/5 < r \leq R_{95}$ measurement interval. On the 16,385-point radial profile this gives

$$\lambda = 2.385686863 \text{ kpc}, \quad N = 3.433448083. \quad (62)$$

The downstream Curvature Structure Mode Index (CSMI) interval is

$$3.30 < N \leq 3.55, \quad (63)$$

so the CSMI structural-class label is “Bulged Spiral.” This is the name of the Structural Mode interval and is not a morphological assertion that the high-redshift quiescent galaxy is a present-day spiral.

A continuum total-variation calculation on the same radial solution gives

$$N_{\text{TV}} = 3.434244641, \quad (64)$$

only 7.97×10^{-4} from the sampled SPARC-equivalent value. Increasing the sampled radial grid from 8193 to 16385 points changes N by 7.95×10^{-4} .

The adopted interval begins at

$$R_{95}/5 = 1.638226 \text{ kpc}, \quad (65)$$

slightly beyond the largest directly resolved structural radius, 1.441132 kpc. The reported Structural Mode is therefore a model-continued full-galaxy readout, not a direct kinematic measurement at R_{95} .

Quantity	Present resolved calculation	Earlier ZF-COS-20115 calculation
R_{95} (kpc)	8.19113	5.80731
λ (kpc)	2.38569	1.68288
Structural Mode N	3.43345	3.45081
CSMI interval	Bulged Spiral	Bulged Spiral
$\Delta N/N_{\text{earlier}}$	−0.50%	—

Table 5: Structural continuity with the earlier effective-radius/oblate-source analysis [6]. The source and dynamical reconstruction are substantially different, while the continuous Structural Mode changes by only about one half percent.

The close agreement in Table 5 is not used to tune the present solution. It is a downstream comparison between two independently structured reductions of the same galaxy. With the field construction, orbit compatibility, and structural readout now separated, the principal physical results can be summarized cleanly.

11 Results and Interpretation

The resolved calculation yields five distinct results connecting the measured stellar support, the reconstructed source, the conservative field, the stationary orbit solution, and the downstream structural mode.

First, the 26 measured stellar regions define 26 local structural states. In the resolved reduction adopted here, each bin’s V_{rms}^2/R support coordinate is used as its local $a_{\star,i}$ and is propagated through the same CPTG galaxy response law. This is important because no single galaxy-wide acceleration is introduced to absorb the spatial variation of the kinematics. The resolved structure is retained rather than averaged away.

Second, those local structural measurements are combined with a baryonic source reconstructed independently from the stellar light. The CPTG response changes the magnitude of the local baryonic field according to the polarization–transport law while retaining the source-field direction. The resulting 26 vectors are simultaneously realized by one conservative three-dimensional potential to substantially better precision than the numerical field-realization criterion. Thus the local solutions are not left as disconnected pointwise calculations; they can be embedded in one gravitational field suitable for dynamics.

Third, the conservative field supports a stationary nonnegative orbit population compatible with all measured resolved line-of-sight velocities and second moments at the strictest tabulated 1 percent tracer tolerance. All 4096 trajectories satisfy the energy qualification criterion. The strongest velocity and second-moment residuals remain below 3σ , including

bin 21, which is retained normally throughout the analysis. This is the central dynamical closure result of the paper.

Fourth, the matched baryonic-only control does not reproduce that closure. At the same 1 percent tracer tolerance its maximum second-moment pull is 3.647σ , with only 24 of 26 V_{rms} bins below 3σ , and its V_{rms} RMSE is 105.35 km s^{-1} compared with 75.30 km s^{-1} for CPTG. Relaxing the numerical tracer tolerance through 10 percent still leaves a maximum baryonic-only V_{rms} pull of 3.072σ . Because the orbit machinery and observation operator are held fixed, this control isolates the effect of replacing the baryonic-only gravitational field with the CPTG field within the specified conditional solve.

Fifth, the full radial continuation gives

$$N = 3.43345. \quad (66)$$

Despite the change from an integrated one-radius source reduction to a resolved local- a_* field with nonaxisymmetric orbit closure, the Structural Mode differs from the earlier $N_{\text{obl}} = 3.45081$ result by only about 0.50%. The agreement is notable because N is calculated after the field solution and was not used to choose the local accelerations, source geometry, conservative interpolation, or orbit weights. It therefore provides a downstream structural continuity check between two different reductions of the same object; because both values remain model-derived from the same galaxy, their agreement is descriptive continuity rather than an independent validation datum.

The high redshift gives the result an additional use. ZF-COS-20115 is observed at an early CPTG native phase, yet the bound-system galaxy law is not modified by an explicit redshift-dependent acceleration factor. The systemic redshift establishes the observed phase, while the registered imaging and kinematic products supply the physical source coordinates used by the numerical reconstruction. The local structural accelerations still come from the galaxy’s own stellar support, and Eqs. (7)–(11) are applied in the same form used for lower-redshift bound systems. In this sense the galaxy provides an early-epoch application of the same CPTG galaxy reduction rather than a separately calibrated high-redshift branch.

12 Limitations and Claim Boundary

The interpretation must remain matched to how the inputs are used. The 26 resolved kinematic measurements enter twice: first through $a_{*,i} = (V_i^2 + \sigma_i^2)/R_i$, and second through the nonnegative orbit-weight solve. The final residuals therefore establish conditional compatibility and stationary existence in the constructed CPTG field. They are not out-of-sample predictions of the same 26 measurements, and the reported pulls are marginal per-bin diagnostics rather than a joint covariance goodness-of-fit statistic. The all-bin 3σ threshold is a feasibility criterion for this conditional stationary solution; it is not a model-selection significance, a posterior probability, or a probability that CPTG is correct.

The central-value orbit closure also does not constitute a full uncertainty-marginalized inference. The measured V_{rms} errors imply median fractional 1σ uncertainty 0.314 in the local $a_{*,i}$ values. Propagating those errors through the local CPTG law alone produces

a median 68-percent fractional half-width 0.0788 in $g_{\text{CPTG},i}$, with a maximum 0.194. The present calculation does not resample those structural inputs through the Hermite reconstruction and the 4096-orbit weight solve, nor does it include covariance among the resolved kinematic bins. The quoted orbit pulls therefore describe the central measured structural state under the specified reduction.

The structural measurement also depends on the adopted $q_{\text{str}} = 0.70$ radius convention. The explicit sensitivity calculation shows changes in $a_{*,i}$ of order ten percent for the source-axis-ratio alternative and substantially larger shifts for some bins under a circular-radius convention. The present closure therefore pertains to the specified support geometry.

The three-dimensional baryonic source is likewise conditional on the adopted lifting. The intrinsic axis ratio is set equal to the projected source axis ratio and the projected first-harmonic factor is carried unchanged along the line of sight. No sensitivity family over intrinsic thickness or alternative line-of-sight harmonic lifts is solved here. The stationary orbit result therefore pertains to this specified three-dimensional source realization.

The continuous conservative field is conditional on the Hermite-kernel reconstruction. Exact agreement at the 26 centers is imposed by construction, while the leave-one-center-out test—including a 165.1° direction error for held-out bin 24—shows that the sparse centers do not uniquely determine the field between them. The evidence archive supplies the Hermite widths and coefficients, evaluates the resulting scalar potential, and verifies its field consistency, but it does not independently regenerate the choice of kernel widths and coefficient construction from the 26 anchors through a separate reconstruction algorithm. The present paper therefore makes no claim that alternate interpolation kernels or alternate reconstruction rules must produce the same detailed orbit closure.

The 4096-orbit library is fully qualified with respect to trajectory integration, energy conservation, and temporal sampling, but the analysis does not establish an asymptotic limit with respect to orbit-basis size. The stationary-existence result therefore pertains to the sampled 4096-orbit library and should not be interpreted as a uniqueness statement for the underlying distribution function.

The evidence archive regenerates the orbit trajectories and moment matrices from the supplied 4096 initial phase-space states, but it does not regenerate the selection of those initial conditions from a potential-independent sampling prescription. Reusing the identical initial states in the CPTG and baryonic-only potentials makes the comparison a deliberately matched basis test, but it does not by itself establish orbit-basis neutrality. The comparative closure statements therefore pertain to this common sampled phase-space basis.

The Structural Mode readout is downstream but model-continued beyond the directly resolved footprint. Its sampled estimator and continuum cross-check are numerically converged on the supplied full- R_{95} profile, but the publication archive does not independently regenerate the entire outer Jeans continuation from detector-level or photometric inputs. A direct stellar-kinematic measurement at R_{95} is not claimed.

ZF-COS-20115 is also observed as part of the Jekyll–Hyde system [4, 1]. The present potential models the reconstructed stellar source of Jekyll and asks whether that field admits a stationary internal orbit population; it is not a time-dependent simulation of the

Jekyll–Hyde interaction or of an external tidal field. The stationary-existence conclusion is therefore conditional on this bound-system representation of Jekyll at the observed epoch.

Finally, the numerical evidence archive begins from processed resolved-bin kinematics and exact spaxel support. Raw detector-level extraction is observational provenance outside the field-to-orbit evidence package. This separation does not alter the dynamical calculation but defines the reproducibility boundary of the accompanying archive. These limitations define the scope of the central result: the resolved CPTG targets admit a conservative realization and a stationary nonnegative stellar population consistent with the measured first and second moments.

13 Conclusion

The earlier ZF-COS-20115 study asked what could be learned from one integrated pressure-supported measurement. The resolved JWST data now permit the stronger question: can the galaxy be treated locally, assembled into one nonaxisymmetric CPTG field, and supported by a stationary stellar population without replacing its resolved structure by a single fitted acceleration scale?

Within the specified resolved reduction, the 26 local structural measurements admit a common conservative CPTG realization and stationary orbit solution. Each resolved stellar region supplies its own structural acceleration. The independently reconstructed baryonic source and the standard CPTG polarization–transport law convert those measurements into 26 local gravitational vectors, and one conservative three-dimensional potential realizes them with maximum fractional mismatch 3.66×10^{-12} . A 4096-orbit library in that potential is fully energy qualified. At the strictest tabulated 1 percent tracer tolerance, a nonnegative stationary orbit population places every measured V and V_{rms} value within the specified marginal per-bin 3σ closure criterion, with maximum pulls 2.431σ and 2.879σ . The matched baryonic-only control, run with the same 4096 initial conditions and downstream observation machinery, reaches a maximum V_{rms} pull of 3.647σ at the same tolerance and remains above 3σ throughout the tested 1–10 percent tracer-tolerance ladder. Its 1 percent V_{rms} RMSE is 105.35 km s^{-1} , compared with 75.30 km s^{-1} for CPTG. Because the same resolved kinematics help define the local structural states and the final orbit weights, these are conditional field-to-orbit compatibility comparisons rather than blind predictions of those measurements.

The downstream full-galaxy Structural Mode is

$$N = 3.433448083, \tag{67}$$

with $\lambda = 2.385686863 \text{ kpc}$ and $R_{95} = 8.191131986 \text{ kpc}$. It remains in the Bulged-Spiral CSMI interval and differs by only about 0.50% from the earlier ZF-COS-20115 oblate-source result. The continuity of N is obtained despite a substantially richer source and dynamical reconstruction and without using N as an input to the fit.

Finally, the galaxy’s redshift has a precise but limited role. It places ZF-COS-20115 at the observed early CPTG phase; the processed imaging and kinematic products supply

the fixed angular-to-physical registration used by the field-to-orbit calculation. Neither operation imposes a redshift scaling on a_* . The gravitational structure used in the resolved calculation is therefore measured from the galaxy at that epoch and passed through the same CPTG galaxy law, providing an early-universe application of the bound-system reduction.

Data, Code, and Evidence Availability

The observational context and spatially resolved JWST/NIRSpec spectroscopy of Jekyll are described by Pérez-González et al. [1]. Public JWST observations provide the observational basis, while the field-to-orbit calculation begins from processed stellar-kinematic, spaxel-assignment, and imaging/source products. The numerical evidence archive accompanying this paper contains the 26-bin structural measurements, the exact 141-spaxel observation support used by the calculation, the reconstructed source components and 3-kpc aperture-mass check, the CPTG and matched baryonic-only conservative-field constructions, 4096 supplied orbit initial conditions, CPTG and baryonic-only orbit moment matrices and weight solutions, structural-radius and local-input uncertainty diagnostics, the sub-percent tracer-closure diagnostic, the full- R_{95} Structural Mode profile, and verification scripts.

The publication evidence archive is

`ZF_COS_20115_Resolved_Nonaxisymmetric_CPTG_Field_to_Orbit_Closure_Evidence.zip`

with SHA-256

`7eae3d99eedc0b9483e71d0300f7491eacc9698ce0a077d7b5c426815b3f5023`

The archive establishes the numerical scope stated in this paper from those processed inputs forward. It does not reproduce the resolved stellar-kinematic extraction from detector-level JWST exposures, nor does it regenerate the upstream selection of the 4096 orbit initial conditions or the complete outer Structural-Mode continuation from raw observational products. These boundaries are stated explicitly in Sec. 12.

Supporting CPTG theory resources are maintained at: <https://github.com/CLG2025/CPTG>

References

- [1] P. G. Pérez-González, F. D'Eugenio, B. Rodríguez del Pino, M. Perna, H. Übler, R. Maiolino, S. Arribas, G. Cresci, I. Lamperti, A. J. Bunker, S. Carniani, S. Charlot, C. J. Willott, T. Böker, E. Parlanti, J. Scholtz, G. Venturi, G. Barro, L. Costantin, I. Martín-Navarro, J. S. Dunlop, and D. Magee, *Accelerated quenching and chemical enhancement of massive galaxies in a $z \simeq 4$ gas-rich halo*, *Nature Astronomy* **9**, 1240–1255 (2025), DOI: 10.1038/s41550-025-02586-8.
- [2] J. Esdaile, K. Glazebrook, I. Labbé, E. Taylor, C. Schreiber, T. Nanayakkara, G. G. Kacprzak, P. A. Oesch, K.-V. H. Tran, C. Papovich, L. Spitler, and C. M. S. Straatman, *Consistent dynamical and stellar masses with a potential light IMF in massive quiescent galaxies at $3 < z < 4$ using velocity dispersion measurements with MOSFIRE*, *Astrophys. J. Lett.* **908**, L35 (2021), DOI: 10.3847/2041-8213/abe11e.
- [3] K. Glazebrook, C. Schreiber, I. Labbé, T. Nanayakkara, G. G. Kacprzak, P. A. Oesch, C. Papovich, L. R. Spitler, C. M. S. Straatman, K.-V. H. Tran, and T. Yuan, *A massive, quiescent galaxy at a redshift of 3.717*, *Nature* **544**, 71–74 (2017), DOI: 10.1038/nature21680.
- [4] C. Schreiber, I. Labbé, K. Glazebrook, G. Bekiaris, C. Papovich, T. Costa, D. Elbaz, G. G. Kacprzak, T. Nanayakkara, P. Oesch, M. Pannella, L. Spitler, C. Straatman, K.-V. Tran, and T. Wang, *Jekyll & Hyde: quiescence and extreme obscuration in a pair of massive galaxies 1.5 Gyr after the Big Bang*, *Astron. Astrophys.* **611**, A22 (2018), DOI: 10.1051/0004-6361/201731917.
- [5] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, *Astrophys. J.* **270**, 365–370 (1983), DOI: 10.1086/161130.
- [6] Carter L Glass Jr, *CPTG Native-Phase Structural Test of ZF-COS-20115: Effective-Radius Closure and Structural Mode N*, (2026), DOI: 10.5281/zenodo.22443339.
- [7] Carter L Glass Jr, *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, (2026), DOI: 10.5281/zenodo.22441039.
- [8] Carter L Glass Jr, *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates*, (2026), DOI: 10.5281/zenodo.22426857.
- [9] Carter L Glass Jr, *Calculating CPTG Galaxy Rotation Curves: DDO-154, NGC-3198, and NGC-7814*, (2026), DOI: 10.5281/zenodo.22418189.
- [10] Carter L Glass Jr, *Structural Mode N in CPTG: Galaxy Structural Classification from the Solved CPTG Field*, (2026), DOI: 10.5281/zenodo.22436682.
- [11] M. Schwarzschild, *A numerical model for a triaxial stellar system in dynamical equilibrium*, *Astrophys. J.* **232**, 236–247 (1979).
- [12] J. Binney and S. Tremaine, *Galactic Dynamics*, 2nd ed. (Princeton University Press, Princeton, 2008).